Rui Qi

Robotics and interactive simulation

+1 801-913-3814 | markqirui@gmail.com | GitHub | LinkedIn

Robotics portfolio | Game portfolio

Education

University of Utah | B.S. in Games

Aug 2022 - Apr 2026 | Degree conferred April 30, 2026 | GPA 3.880/4.000

Selected coursework: Programming for All 1 and 2; Introductory Statistical Inference; Game Engine Fundamentals; Traditional Game Development. Dean's List, multiple semesters.

Robotics Project

Visual Grasp Verification and Failure Recovery | Independent project | Oct 2026

- Defined task goals, tested demonstrations and reported missed grasps, empty transport and failed retries. Questioned an unnecessary obstacle push, prompting alternative-grasp comparisons and a stacked-object task.
- The simulation combines color/depth localisation, programmed grasping and extraction, visual payload checks, and recovery after acquiring the wrong object. Current recovery logic is programmed, not a newly trained policy.
- Current workflow and a relocate-first baseline each completed 7/7 development layouts. Of six stacked cases, two used direct extraction and four required recovery; mean task times were 14.64 vs 14.32 simulated seconds. No untouched final test yet.
- A separate small neural policy trained with PPO for 96 episodes selected among three grasp strategies. On 12 new development-validation layouts, policy and rules each completed 10/12; the policy was slower (12.51 vs 11.03 simulated seconds).

Role: task direction, hands-on failure feedback, comparison requirements and result review. Implementation, debugging and experiment automation used extensive AI assistance.

Relevant Experience

Qainbow LLC | Technical / Systems Designer and Producer, founding team

Utah, USA | Aug 2026 - Present

- Translate design goals into implementation tasks and coordinate iteration. For Breach Brew, develop reusable Blueprint systems, parameter-driven environmental materials and Niagara feedback.

TimeStay | Part-time Systems and Balance Designer

Shanghai, China | 2026 - Present

- Contribute gameplay rules, numerical configuration and balance content; communicate implementation needs with the development team.

Selected Technical Projects

Punk Plush Panic | Unity capstone | 2025 - 2026

Improved boss attack timing and telegraphs; built configurable portals and path-driven traps; iterated from playtests and reduced scope for delivery. Released on Steam.

The E Book | Unity | Commercial project

Maintained gameplay-configuration spreadsheets and redesigned the data workflow; developed an AI-assisted Excel-to-Unity conversion tool.

Skills

Technical: Unreal Blueprint; basic Python and JSON; Excel configuration; Git and Perforce.

Simulation and content: project experience with Isaac Sim / Isaac Lab and RGB-D workflows; Unity, Godot, Niagara and material parameters; basic Maya / Blender modelling.

Languages: native Chinese; fluent English.